

# Alessio Boetti

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**Gender:** Male **Date of birth:** 1999 **Nationality:** Italian

## ABOUT ME

**AI Engineer & Data Scientist** with solid experience designing, implementing and deploying data pipelines and end-to-end Machine Learning and Generative AI systems in regulated enterprise-grade cloud and on-premise environments. Built scalable solutions for fraud detection, forecasting, NLP, and LLMs to guide strategic decisions and optimize business performance, across the entire infrastructure, data, and model lifecycle. Focused on mission-critical contexts where reliability, software quality, and business impact are key drivers.

## WORK EXPERIENCE

**Proxima Group** [www.proximagroup.eu](http://www.proximagroup.eu)

**City:** Rome (Remote) | **Country:** Italy

[ 06/10/2025 - Current ]

### Consultant Analytics Intelligence

Contributed, for a market-leading FinTech company, to the enterprise-quality development of a new advanced **banking balance sheet simulation** component within an Enterprise Risk Management (ERM) platform, used by banking institutes to perform financial modeling, anticipate financial trends, simulate geopolitical and socioeconomic scenarios, optimize risk management, and support strategic decision-making.

- Backend development of data modeling classes and REST APIs with FastAPI and Pydantic, supporting the financial simulation logic.
- Unit testing (Pytest), functional validation, and integration of every new development or change, in collaboration with functional analysts and quants, to ensure the correctness and reliability of the computational models.
- Data manipulation and transformation with pandas to support the simulation component.

**TAS S.p.A.** [www.tasgroup.eu](http://www.tasgroup.eu)

**City:** Casalecchio di Reno, Bologna (Remote) | **Country:** Italy

[ 31/10/2022 - 03/10/2025 ]

### Banking Data Scientist & Machine Learning Engineer

**Summary:** Responsible for the design, development, release, and evolutionary maintenance of end-to-end AI and Machine Learning solutions for the banking sector, following the entire project lifecycle: requirements analysis, solution design, development, validation, deployment, monitoring, and operational support. Key areas: data pipelines & ETL/ELT, fraud detection, forecasting, LLM- and GenAI-based chatbots, business process automation, and Business Intelligence.

Development and full lifecycle management of:

- **4 enterprise platforms for banking fraud detection** on AWS infrastructure using **Apache Spark/PySpark, Terraform, Docker, and Kubernetes** significantly improving fraud detection accuracy and reducing clients' financial losses. Projects: predictive models with Explainable AI (XAI) and monitoring/drift detection for

retraining, automatic anti-fraud rule generation systems, chatbot for customer anti-fraud screening, and Business Intelligence dashboards. Some solutions were reviewed through **formal audit** processes and successfully passed them, and contributed to identifying over **€25,000** in fraud.

- **Liquidity forecasting engine:** forecasting treasury liquidity for banking institutions, supporting financial planning and optimizing liquidity management through time series analysis and statistical techniques.
- **GenAI-based chatbots:** solutions integrated with corporate knowledge bases and internal systems for B2B use cases, customer support, and operator assistance, leveraging LLMs and NLP.
- **Anomaly detection for IT infrastructure monitoring:** continuous analysis of system logs to automatically identify anomalies and malfunctions, reducing detection times and service downtime.
- **Statistical data analysis and reporting** on bank customer demographics and behavior (Business Intelligence).

**Process optimization:** automated testing cycles and manual infrastructure processes, significantly reducing operational time.

**Development and management of CI/CD pipelines and DevOps/MLOps workflows** to ensure reliable, reproducible, and scalable releases using Docker, Kubernetes, Rancher, Argo, GitHub/GitLab, Pytest, Kaniko, Coverity, SonarQube, and Black Duck. Infrastructure-as-code management (Terraform), access control (Keycloak, Duo, AWS IAM), and database schema versioning with Liquibase.

**Technical mentoring and AI culture promotion:** designed and delivered an internal course on Artificial Intelligence and Machine Learning for senior colleagues, including theoretical deep-dives, business use cases, and hands-on demonstrations in Python.

## EDUCATION & TRAINING

[ 22/09/2021 - 03/10/2025 ]

### Master's Degree in Artificial Intelligence (Information Engineering, LM-32)

**University of Florence** <https://www.unifi.it/en>

**City:** Florence | **Country:** Italy | **Final grade:** 110 | **Level in EQF:** 7

#### **Thesis: Graph Neural Networks for Banking Fraud Detection**

Projects: implemented 2 deep learning models (U-Net, DDPM) to detect anomalous images (defective industrial products), a machine learning model to detect fraudulent banking transactions, multiple deep learning models to correctly classify images (CNN, ResNet), a deep learning model to resist adversarial attacks on images (CNN), built and trained an LLM from scratch for text generation (GPT), trained an LLM for text classification (BERT), applied GNN to document analysis.

Statistics, Generative AI, Deep Learning, Neural Networks, Machine Learning, Natural Language Processing (NLP), Large Language Models (LLM), Transformers, Computer Vision, Adversarial Learning, Anomaly Detection, Out-of-Distribution Detection, Big Data, Innovation Management, Network Security, Robotics.

[ 31/08/2022 - 31/10/2022 ]

### Academy "Banking Data Scientist"

**Umana Forma S.r.l & UMANA S.p.A.**

Course on AWS, R, Python, Data Science, Big Data, Apache Spark & PySpark, SQL & NoSQL Databases.

Final project on banking fraud detection using Machine Learning models.

[ 31/08/2017 - 30/04/2021 ]

### Bachelor's Degree in Electronics and Telecommunications Engineering (Telecommunications course, Information Engineering, L-8)

**University of Florence** <https://www.unifi.it/en>

**City:** Florence | **Country:** Italy | **Level in EQF:** 6

**Thesis:** Design of a congestion control protocol in TCP/IP networks through machine learning algorithms

Mathematics, Physics, Electronics, Electrical Engineering, Telecommunications

[ 30/11/2016 ]

**High School Diploma (Secondary Education Certificate)**

***I.S.S. Liceo Scientifico A. M. Enriques Agnoletti***

**City:** Sesto Fiorentino, Florence | **Country:** Italy | **Level in EQF:** 4

## TECHNICAL SKILLS

### Areas of expertise

Machine Learning, Deep Learning, Statistics, Generative AI (LLM, RAG, Fine-tuning, Agentic AI, Embeddings, Vector databases, Vector stores), Natural Language Processing (NLP), Computer Vision, Fraud Detection, Anomaly Detection, Tabular Data, Graph Data, Graph Neural Networks, Time Series Analysis, Adversarial Learning, Supervised/Unsupervised/Self-Supervised Learning, Explainable AI (XAI), Transformers, CNN, RNN, LSTM, MLOps, Data Engineering, Distributed Computing, Numeric Optimization, Serverless Architectures, Microservices Architectures.

Languages: Python, R, SQL

Cloud & Distributed computing: AWS (S3, DynamoDB, SageMaker, S3, Lambda, Glue, QuickSight, API Gateway, IAM, CloudWatch, etc.), Azure, Apache Spark & PySpark

Containerization & Orchestration: Docker, Docker Compose, Kubernetes, Rancher, Argo, K9S

CI/CD & DevOps: Git/GitHub/GitLab, Terraform, Azure DevOps, Coverity, SonarQube, Black Duck, Kaniko

Databases: PostgreSQL, MongoDB, SQLAlchemy, Liquibase

Web Development: JWT, SSO & OAuth2, Keycloak & Duo, Flask, FastAPI, Django, Streamlit, Gradio

### Libraries

Data Analysis: Pandas, NumPy, Scipy, Matplotlib, Plotly, Seaborn, etc.

Machine Learning: Scikit-Learn, XGBoost, CatBoost, Statsmodels, Nixtla

AI & GenAI: PyTorch, LangChain, OpenAI SDK

Graphs: PyTorch Geometric, Deep Graph Library, NetworkX

Optimization: HyperOpt, Optuna

Asynchronous Tasks & Queues: Celery, RabbitMQ

Other: Pytest, Pydantic, Ruff

## LANGUAGE SKILLS

**Mother tongue(s):** Italian

**Other language(s):**

**English**

**LISTENING:** C1 **READING:** C1 **WRITING:** C1

**SPOKEN PRODUCTION:** B2 **SPOKEN INTERACTION:** B2

**French**

**LISTENING:** A1 **READING:** A1 **WRITING:** A1

**SPOKEN PRODUCTION:** A1 **SPOKEN INTERACTION:** A1

## SKILLS

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Digital Marketing | SEO | WordPress | ECDL | Google Analytics | Social Network | Email Marketing | C++ | Python | R | Microsoft Office

## SOFT SKILLS

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### Management

Ability to manage stress, time, and priorities.

### Leadership

Coordination, teamwork, leadership, decision-making, problem-solving, and team management skills developed as a team leader on projects at TAS S.p.A.

### Public Speaking

Public speaking skills developed through a theater course.

## CERTIFICATIONS

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**AWS Certified Machine Learning Specialty, 2023, Udemy**

**Customer Analytics in Python, 2023, Udemy**

**Introduction to Git for GitLab projects, 2023, Udemy**

**ISO 56000 Innovation Management, 2021, University of Florence**

**B2 Cambridge English, 2016, British Institute of Florence**

**FastAPI - The Complete Course 2026, Udemy**

**MongoDB Python Developer Path, 2025, MongoDB**

**PySpark Essential Training Introduction to Building Data Pipelines, 2026, LinkedIn Learning**

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Sesto Fiorentino, Florence, 23/07/2026



Alessio Boetti